

Literature status: the compact-symmetry spectral-inequality problem

Automated mathematics run `fa_banach_001`, lane 2

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Status: literature already answered, affirmative for open problem 2 under global matrix-symbol positivity.

Source question

Duván Cardona, *Spectral inequalities for elliptic pseudo-differential operators on closed manifolds*, arXiv:2209.10690, Section 3, physical page 18, second bullet. The paper asks for which manifolds the coordinate-invariance restriction $\rho \geq 1 - \delta$ can be removed from Theorem 1.4 and conjectures that, on a manifold with symmetries and under a suitable positivity condition on the symbol, its spectral inequality remains valid for every $0 \leq \delta < \rho \leq 1$.

Supporting answer

Duván Cardona, Julio Delgado, and Michael Ruzhansky, *Estimates for sums of eigenfunctions of elliptic pseudo-differential operators on compact Lie groups*, arXiv:2209.12092. On physical page 2 the introduction states that validity in the complete range $0 \leq \delta < \rho \leq 1$ was an open problem prior to that work. Theorem 1.1, starting on physical page 3, takes a compact Lie group G and a positive elliptic operator

$$A \in \Psi_{\rho,\delta}^m(G \times \widehat{G}), \quad 0 \leq \delta < \rho \leq 1,$$

assumes its global matrix-valued symbol satisfies $\sigma_A(x, \xi) \geq 0$, and proves for every nonempty open $\omega \subset G$ that

$$\|v\|_{L^2(G)} \leq C_1 e^{C_2 \lambda} \|v\|_{L^2(\omega)} \quad (v \in \text{span}\{e_j : \lambda_j \leq \lambda\}).$$

This is precisely the compact-symmetry realization requested by the source, with the previously unspecified positivity condition made explicit.

Provenance and scope

The supporting authors knew the problem: the supporting paper shares the source author, cites Cardona's source preprint, and explicitly calls the same complete-range statement previously open. Therefore the provenance is `literature_already_answered`, not an agent-inferred implication.

Only the second source bullet is resolved. The first question, on suitable boundary problems and in particular higher polyharmonic powers with boundary, is outside this identification. The global matrix-symbol condition is also stronger than positivity of the principal symbol alone.

Evidence

The packet contains exact local compilations of the stored arXiv sources as `source_paper.pdf` and `supporting_paper_2209.12092.pdf`. The source statement is on physical page 18; the decisive supporting introduction and theorem begin on physical pages 2 and 3.